

Multi-Branch Forensic Architecture for ID Card Presentation Attack Detection with Portrait Extraction

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Abstract

Due to the continuing digitalization of our lives, the need for a secure digital system to verify someone's identity has never been higher. With this in mind, attacks on these systems have also improved and multiplied. This paper presents our solution, submitted to the Second Competition on Presentation Attack Detection on ID Cards (IJCB 2025), which adopts a single-model detection approach to explore its potential for multiple tasks. This approach first uses a RetinaFace-based portrait extraction system that adaptively crops ID card images to capture essential forensic features while maintaining sufficient context. After the portrait is extracted, a multi-branch convolutional neural network that integrates backbone features with specialized texture and edge analysis branches checks for attacks. It also addresses the challenges posed by the increasing sophistication of presentation attacks, including print, screen, and composite manipulations. Experimental validation shows the effectiveness of combining preprocessing with forensically-aware feature extraction for reliable presentation attack detection.

1 Introduction

Rapid digitization has reshaped identity verification across banking, healthcare, and border control, shifting from in-person checks to remote workflows and exposing new attack surfaces. Presentation Attack Detection (PAD) is now essential for separating bona fide presentations from increasingly sophisticated forgeries, including high-quality prints, screen replays, and composite edits. Yet many PAD systems rely on global document cues, underuse facial regions, and struggle with limited data and poor cross-document generalization.

We address these gaps with a two-stage approach that couples targeted preprocessing with a forensically aware deep model, tailored to the IJCB PAD-ID Card 2025 competition. The competition aims to benchmark methods under a common protocol and identify effective models via unbiased evaluation. It features two tracks [1]: (1) a private dataset of template-generated PVC documents as bona fide samples with print, screen, and composite attacks (see Fig. 2); and (2) experiments on last year's public and custom datasets. This paper focuses on Track 1.

2 Related Work

Early presentation attack detection approaches relied on handcrafted features and traditional machine learning classifiers [4]. The introduction of convolutional neural networks (CNNs) significantly advanced the state of the art, with Arlazarov et al. [5] demonstrating CNN effectiveness on mobile device-based document analysis. Recent competition results [6] show that ensemble methods and multi-modal approaches achieve superior performance compared to single-model solutions.

Texture analysis has emerged as a crucial component for distinguishing bona fide from attacked presentations. Print attacks introduce characteristic halftone patterns and artifacts detectable through specialized analysis, while screen attacks exhibit moiré patterns and RGB subpixel effects [7]. Composite attacks involving facial manipulation pose unique challenges requiring sophisticated edge and discontinuity analysis [8].

Multi-branch feature extraction has shown promise in forensic applications, enabling systems to capture complementary information from different imaging and attack modalities. These architectures allow simultaneous analysis of high-level semantic features, texture patterns, and edge characteristics, leading to more robust detection across diverse attack scenarios [12].

3 Methodology

Our approach consists of two main stages: portrait extraction using AI that preserves forensically relevant information, followed by a multi-branch neural network combining semantic understanding with specialized forensic analysis. For training the system processes input ID card images through portrait extraction to form a modified dataset ready for final classification, with multiple complementary branches optimized for detecting specific attack artifacts.

3.1 RetinaFace-Based Portrait Extraction

Traditional face detection methods often crop too tightly around the detected face, removing contextual information that may contain crucial attack artifacts. The proposed extraction system implements adaptive cropping to preserve forensically relevant context while maintaining consistent input dimensions.

Algorithm 1 outlines our multi-fallback detection strategy for portrait extraction. The primary detection stage uses RetinaFace [9] with a confidence threshold of $\tau = 0.7$. If the primary detection fails, the system progressively applies CLAHE enhancement, Haar cascade detection, and template-based region estimation as fallback stages.

Algorithm 1 Multi-Fallback Portrait Extraction

```

1: Input: ID card image  $I$ 
2: Output: Cropped portrait region  $P$ 
3: Apply RetinaFace detection ( $\tau = 0.7$ )
4: if faces detected then
5:   Select best face based on size and position
6: else
7:   Apply RetinaFace on CLAHE-enhanced image
8:   if faces detected then
9:     Select best face
10:  else
11:    Apply Haar cascade detection
12:    if faces detected then
13:      Select best face
14:    else
15:      Apply template-based estimation
16:    end if
17:  end if
18: end if
19: Crop with padding:  $P = \text{crop}(I, \text{bbox}, 0.8)$ 

```

The adaptive padding mechanism ($\alpha = 0.8$) captures facial areas and surrounding document context containing potential attack artifacts, with automatic expansion to maintain minimum 384×384 pixel resolution.

3.2 Multi-Branch Forensic Architecture

The classification system employs three complementary branches: a ConvNeXt backbone [11] for extracting high-level semantic features, a texture analysis branch for detecting print-related artifacts, and an edge analysis branch for identifying composite attack presentations.

Backbone: The ConvNeXt architecture pretrained on ImageNet provides robust high-level semantic feature extraction. Early layers are frozen to mitigate overfitting, while the parameters of the final stage are fine-tuned for document-specific adaptation.

Texture Branch: Designed to capture print-related artifacts by applying adaptive pooling to a 32×32 resolution, followed by lightweight convolutional layers with batch normalization, and dimensionality reduction to 64 features. This branch is particularly effective in detecting halftone patterns and color bleeding.

Edge Branch: Designed to detect composite attack artifacts by applying specialized downsampling and edge detection kernels, resulting in 32-dimensional feature representations that capture boundary discontinuities.

Feature integration is performed using learnable weights:

$$F_{\text{combined}} = w_1 * F_{\text{backbone}} + w_2 * F_{\text{texture}} + w_3 * F_{\text{edge}}, \quad (1)$$

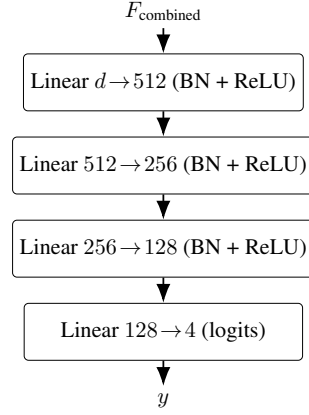


Figure 1: Architecture of the classification head. BN denotes batch normalization and ReLU the rectified linear unit activation. The final linear layer outputs four logits (one per class), forming y .

where $w_1, w_2, w_3 \in \mathbb{R}$ are learnable scalar parameters.

The classification head applies progressive dimensionality reduction (Fig. 1), a standard progressive MLP design widely used in PAD/forgery detection.

3.3 Training Strategy

We employ a dynamic learning rate schedule where each parameter group has its own step size. The learning rate remains unchanged as long as average validation F1-score improves. If no improvement is observed for five consecutive epochs, the rate is reduced by a constant factor. This mechanism prevents premature decreases in learning rate and supports stable convergence during training.

To address class imbalance, conservative weighting is applied with a capped maximum weight ($w_{\text{max}} = 3.0$):

$$w_i = \min \left(\frac{N}{K \cdot n_i}, 3.0 \right), \quad (2)$$

where N is the total number of samples, K the number of classes, and n_i the number of samples in class i .

Finally, forensic-aware augmentation simulates realistic attack scenarios:

- **Screen simulation:** RGB channel shifts, moiré patterns,
- **Print simulation:** noise injection, compression artifacts,
- **Composite artifacts:** edge enhancement,
- **Lighting variations:** brightness and contrast changes.

4 Experiments and Results

4.1 Datasets

The dataset consists of 12,000 ID pictures equally split into four categories: three attack types (print, screen, composite) (Figure 2) and one bona fide class (Table 1).

Class	Bonafide	Print	Screen	Composite
Train	2500	2500	2500	2500
Val	500	500	500	500

Table 1: Data split by class for training and validation sets.

4.2 Training Setup and Implementation Details

Training was performed in PyTorch with an input resolution of 384×384 pixels, a batch size of 32, and up to 250 epochs. Early stopping was triggered after 15 epochs without improvement. Optimization used the AdamW optimizer with multi-scale learning rates: 5×10^{-7} for the backbone, 1×10^{-5} for forensic branches, 1×10^{-4} for feature weights, and 5×10^{-6} for the classifier.

Hardware: Experiments were run on a supercomputer with 16 CPU cores, 128 GB of RAM, and dual NVIDIA A100 GPUs (80 GB memory each).

4.3 Experiments

Multiple `timm` [10] models were evaluated on the cropped dataset. Among them, the ConvNeXt [11] architecture was chosen as it supports a sufficiently small input size without compromising the quality of the cropped images, while also achieving state-of-the-art performance on the ImageNet benchmark.

Multiple branch configurations were tested to achieve optimal detection performance. In the end, a texture branch and an edge branch were selected to address the challenging detection of print and composite attack cards.

The basic success metrics used are APCER (Attack Presentation Classification Error Rate), BPCER (Bonafide Presentation Classification Error Rate) and EER (Equal Error Rate). The metrics are defined as follows:

$$APCER_{PAIS} = 1 - \frac{1}{N_{PAIS}} \sum_{i=1}^{N_{PAIS}} RES_i \quad (3)$$

$$BPCER_{PAIS} = \frac{1}{N_{BF}} \sum_{i=1}^{N_{BF}} RES_i \quad (4)$$

where N_{PAIS} represents the number of attack presentation samples in the dataset (presentation attack instruments), N_{BF} represents the number of genuine examples (bona fide), and RES_i represents how the i -th example was classified. RES_i is 1 if the i -th example is classified as an attack and 0 if it is classified as genuine.

The Equal Error Rate (EER) is defined as the operating point at which the APCER and BPCER are equal. Formally, let τ denote the decision threshold. Then the EER is obtained as

$$EER = APCER(\tau^*) = BPCER(\tau^*), \quad (5)$$

where τ^* is the threshold at which the false acceptance and false rejection rates are balanced.

The competition organizers will provide a more detailed description of the results and contributions received during the competition in an article to be published in IJCB 2025 [3].

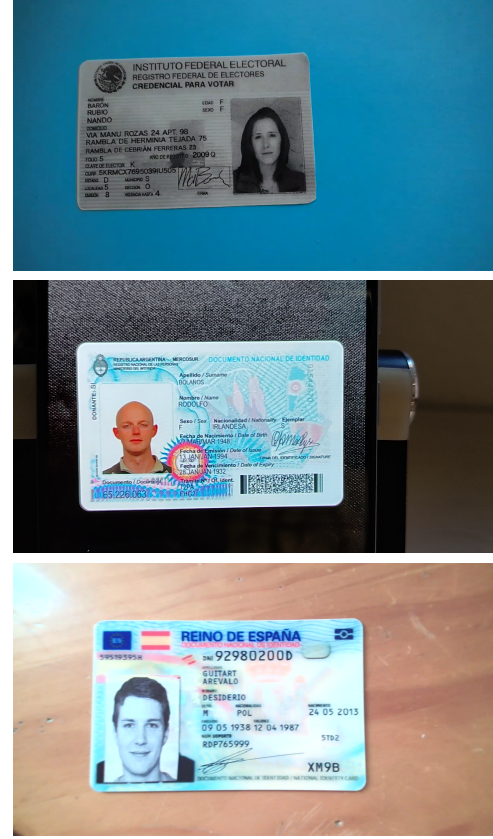


Figure 2: Representative samples: print attack with paper texture artifacts (top), screen attack with moiré patterns (middle), and composite attack with facial manipulation (bottom).

4.4 Evaluation Protocol

Evaluation protocol follows ISO/IEC 30107-3 [2] using EER and BPCER at fixed APCER values: 10% (BP10), 5% (BP20), and 1% (BP100). Final ranking uses:

$$AVR = 0.2 * BP10 + 0.3 * BP20 + 0.5 * BP100 \quad (6)$$

4.5 Portrait Extraction Performance

The RetinaFace-based system (Algorithm 1) achieves an 88.3% success rate on the first attempt and 98.7% with the multi-fallback strategy. The average processing time is 0.23 seconds per image. The adaptive padding mechanism effectively preserves relevant context, leading to improved downstream classification performance compared to tight face crops.

4.6 Validation results

Since competition testing was performed online on a hidden independent test set, all reported performance results are based on the local validation subset.

During training, two success metrics were followed to determine whether the model was achieving the expected results. The first metric was the F1-score, which measured the harmonic mean of precision and recall on the trained model for each category and overall average. On early stopping it achieved:

Approach	Bonafide	Print	Screen	Composite	Avg
With crop	89.83	92.85	97.8	94.88	93.84
Without crop	83.54	84.6	98.76	98.7	91.4

Table 2: F1-score [%] on last epoch by class.

The second metric follows the ISO/IEC 30107-3 format [2] and shows the models achieved success based on it. At early stopping, the model achieved:

Approach	EER	BP10	BP20	BP100	AVR
with crop	5.60	7.80	16.20	51.40	32.12
without crop	8.97	25.40	45.40	75.60	56.50

Table 3: Validation results following the evaluation protocol presented in Section 4.4.

4.7 Competition Performance

The results of the proposed algorithm were below the competition baseline (Track1_Host_Team). Although not outstanding, the approach achieved a respectable 6th place overall (Table 4).

Team	EER	BP10	BP20	BP100	AVR
dragons	11.34	13.21	24.39	61.04	40.48
Idiap	14.12	18.30	30.81	61.43	43.62
Track1_Host_Team	16.51	27.80	45.26	77.11	57.70
IDCH	21.66	39.36	51.66	71.75	59.25
Asmodeus	24.01	43.08	57.28	80.73	66.16
UNI-FRI-FE (Ours)	26.53	51.29	65.13	83.95	71.77
IDVC-PAD-IDCARD	22.41	46.86	65.59	91.33	74.71
VISTeam	36.30	61.88	73.09	89.09	78.95
PADINO-v2	32.97	72.05	84.23	97.04	88.20

Table 4: Competition results following the evaluation protocol presented in Section 4.4.

The most successful entries used document-centric pipelines—explicit card detection/segmentation with homography rectification—paired with large ViT backbones (e.g., CLIP ViT-L/14) adapted via LoRA and a shallow MLP head, outperforming single-stream CNNs.

Portrait cropping slightly reduces the per-class F1-score for Screen and Print (Table 2). However, it yields a better global operating point, and the EER and BPCER at fixed APCER are lower, so the overall AVR improves (Table 3). Together with the approaches of the winning teams (Table 4), this suggests that high-quality preprocessing produces a better training outcome.

5 Discussion and Conclusion

This paper presents a comprehensive approach to presentation attack detection, combining intelligent portrait extraction with forensically-aware deep learning architectures. The RetinaFace-based extraction method achieves a 98.7% success rate while preserving forensic context. The proposed multi-branch architecture effectively integrates semantic, texture, and edge features, achieving strong performance across different attack categories.

Experimental evaluation on the validation set demonstrates detection F1-scores of 93% for print attacks, 98% for screen attacks, and 95% for composite attacks (Table 2), confirming the effectiveness of the proposed approach. The stable training progression indicates efficient learning without overfitting.

One current limitation is that testing was only possible within the scope of the competition, leaving insufficient time to further refine and improve the system. Future work will explore extending this approach to systems incorporating multiple recognition models and diverse branch configurations to further improve results. Additionally, domain adaptation for cross-dataset generalization, attention mechanisms for enhanced feature integration, and advanced augmentation strategies to address emerging attack methodologies are expected to further strengthen the system. Overall, the proposed methodology contributes significantly to advancing presentation attack detection, providing a robust foundation for secure identity verification systems.

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